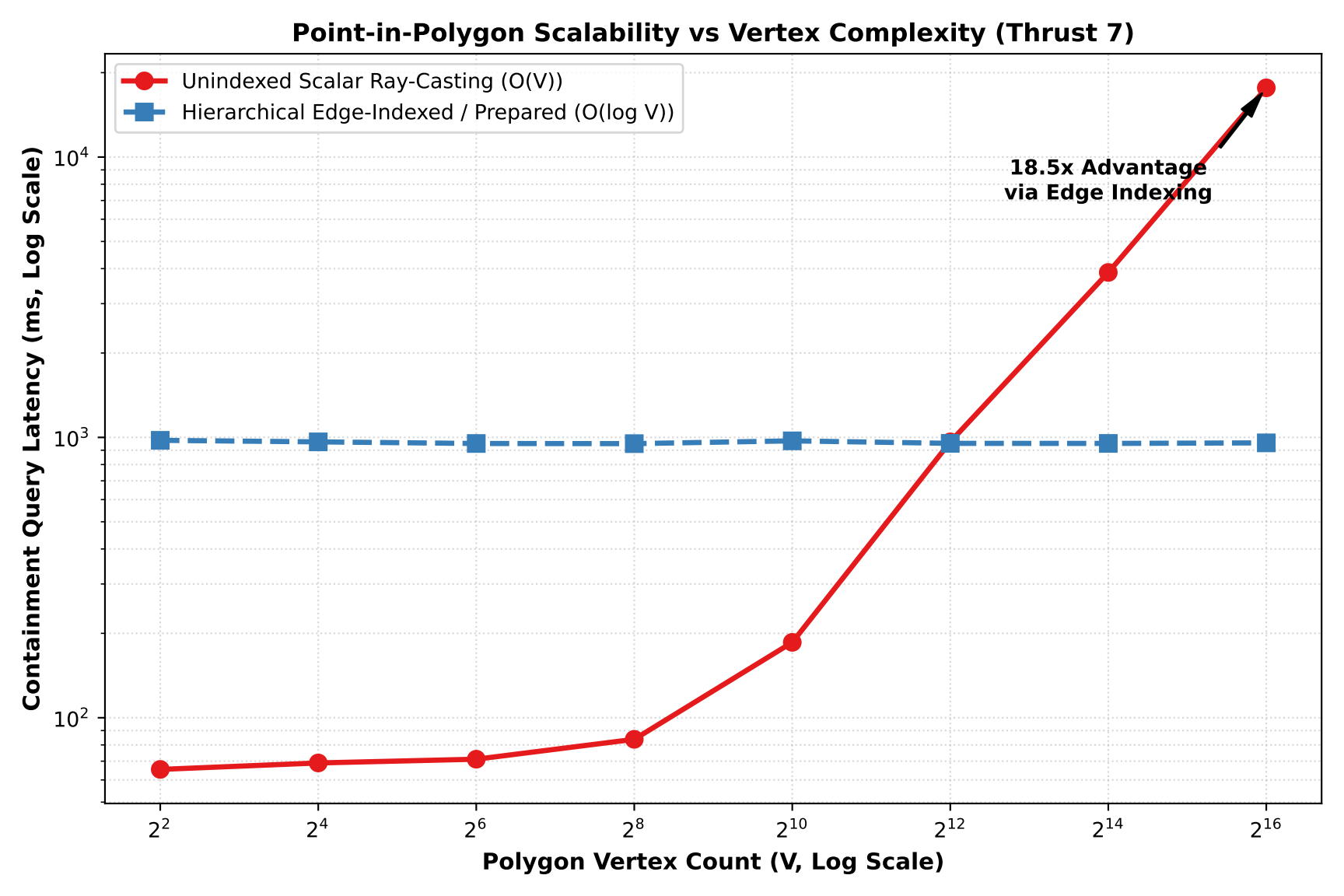
Point-in-Polygon Scalability vs Vertex Complexity (Thrust 7)
Unindexed Scalar Ray-Casting (O(V))
Hierarchical Edge-Indexed / Prepared (O(log V))
18.5x Advantage
via Edge Indexing
Containment Query Latency (ms, Log Scale)
10^4
10^3
10^2
2^2
2^4
2^6
2^8
2^{10}
2^{12}
2^{14}
2^{16}
Polygon Vertex Count (V, Log Scale)